

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

B.Tech. (CL/EC/ME/ CE/EE/IT)

SEM - II University Examination, Nov- 2013

MA - 102 Engineering Mathematics-II

Date: 28.11.13, Thursday

Time: 10.00 a. m to 01.00 p. m.

Maximum Marks: 70

Instructions: (1) All questions are compulsory.

(2) Indicate clearly, the option you attempt along with its respective question number.

(3) Figures to right indicate marks.

SECTION-I

Q-1 (a) Solve $(e^y + 1)\cos x dx + e^y \sin x dy = 0$. [03]

(b) Solve $y'' - 6y' + 5y = \sin 2x$. [04]

Q-2 Solve the following differential equations

(a) Using method of variation of parameters $\frac{d^2y}{dx^2} + 25y = \sec 5x$. [05]

(b) $\frac{dx}{dt} = 7x - y$, $\frac{dy}{dt} = 2x + 5y$. [05]

(c) $(y^3 - 3xy^2)dx + (2x^2y - xy^2)dy = 0$. [04]

OR

Q-2 Solve the following differential equations

(a) $y''' + 3y'' + 3y' + y = 30e^x$. [05]

(b) $x^2y'' + xy' + y = x^2 + 1$. [05]

(c) $\frac{dy}{dx} + y \sec x = \tan x$. [04]

Q-3 Solve the following Partial differential equations

(a) $(mz - ny)p + (nx - lz)q = (ly - mx)$. [05]

(b) (i) $p + q = px + qy$ (ii) $pq = p + q$. [05]

(c) (i) $z = px + qy + \sqrt{1 + p^2 + q^2}$ (ii) $z - px - qy = 2\sqrt{pq}$ [04]

OR

Q-3 Solve the following Partial differential equations

(a) $(y - z)p + z(x - y)q = z - x$. [05]

(b) $z^2(p^2 + q^2 + 1) = 1$. [05]

(c) Form the Partial differential equation

(i) $2z = (x + a)(y + b)$ (ii) $z = xy + f(x^2 + y^2)$. [04]

SECTION-II

Q-4 (a) Find the eigenvalues and corresponding eigenvectors for the matrix

$$A = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 2 \end{bmatrix}. \quad [06]$$

(b) $\int_{-1}^1 P_7(x) P_4(x) dx = \underline{\hspace{2cm}}$ [01]

Q-5 (a) State Cayley Hamilton theorem. Find the characteristic equation of the matrix [06]

$$A = \begin{bmatrix} 1 & 1 \\ 2 & 3 \end{bmatrix} \text{ and hence evaluate the matrix equation}$$

$$A^5 - 4A^4 - 7A^3 + 11A^2 - A - 10I.$$

(b) Change the order of integration in the integral $\int_0^{\infty} \int_x^{\infty} \frac{e^{-y}}{y} dy dx$ and evaluate it [04]

(c) Prove $J_{-1/2}(x) = \sqrt{\frac{2}{\pi x}} \cos x.$ [04]

OR

Q-5 (a) If $A = \begin{bmatrix} -1 & 2+i & 5-3i \\ 2-i & 7 & 5i \\ 5+3i & -5i & 2 \end{bmatrix}$, Show that A is a Hermitian matrix and iA is a [06]

skew Hermitian matrix.

(b) Evaluate $\iint r^3 \sin 2\theta dr d\theta$ over the area bounded in the first quadrant R between the circles $r=2$ and $r=4$. [04]

(c) Express $f(x) = 4x^3 - 3x^2 + 2x + 1$ in terms of Legendre polynomials. [04]

Q-6 (a) Evaluate (i) $\int_0^{\infty} x^2 e^{-x^4} dx$ (ii) $\int_{-\infty}^{\infty} e^{-k^2 x^2} dx$ [06]

(b) Evaluate $\iiint_V z(x^2 + y^2) dx$ over the cylinder $x^2 + y^2 = 1$ intercepted by the plane $z=2$ and $z=3$. [04]

(c) Define Symmetric Matrix and Orthogonal matrix with example. [04]

OR

Q-6 (a) Evaluate (i) $\int_0^2 x^4 (8 - x^3)^{-\frac{1}{3}} dx$ (ii) $\int_0^n x^n (n - x)^p dx$ [06]

(b) Prove that $A = \frac{1}{2} \begin{bmatrix} 1+i & -1+i \\ 1+i & 1-i \end{bmatrix}$ is a Unitary matrix. [04]

(c) Find the area between the parabola $y^2 = 4ax$ and $x^2 = 4ay$. [04]